Throughout the book I'll show you how to configure the services in Polar Bookshop in different ways. I'll start by exploring the options offered by Spring Boot properties and profiles, and when to use them. You'll then learn how to apply external configuration by using environment variables when running a Spring application as a JAR and as a container. Then you'll see how to centralize the configuration management through a configuration server with Spring Cloud Config. Finally, I'll teach you how to use ConfigMaps and Secrets in Kubernetes.

ROUTING

Polar Bookshop, being a distributed system, will need some routing configuration. Kubernetes has a built-in service discovery feature that can help you decouple services from their physical addresses and hostnames. Cloud native applications are scalable, so any interactions between them should take that into account: which instance should you call? Once again, Kubernetes offers you a load-balancing feature natively, so you don't need to implement anything in your application.

Using Spring Cloud Gateway, I'll guide you through implementing a service that will act as an API gateway to shield the outside from any internal API changes. It will also be an edge service that you'll use to address cross-cutting concerns, like security and resilience, in one place. Such a service will be the entry point to Polar Bookshop, and it will have to be highly available, performant, and fault-tolerant.

OBSERVABILITY

The services in the Polar Bookshop system should be observable to be defined as cloud native. I'll show you how to use Spring Boot Actuator to set up health and info endpoints and expose metrics with Micrometer to be